

EXPERIMENT: 3

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SURVEY REPORT ON EMERGING TRENDS IN LOW-CODE APPLICATION DEVELOPMENT

Aim:

To study the emerging trends in low-code application development and understand how low-code platforms can be used to develop applications with minimal traditional programming.

Traditional Application Development vs Emerging Trends in Low-Code Application Development:

2.1 Traditional Application Development:-

Traditional application development involves building applications mainly through manual coding. Developers write code for the interface, business logic, database, testing, and deployment. It provides high flexibility and customization but requires skilled programmers and generally takes more time and effort to develop and maintain.

2.2 Low-Code Application Development:-

Low-code development uses visual tools, drag-and-drop components, pre-built modules, and reusable elements to create applications with minimal manual coding. It supports faster development and allows both professional developers and business users to participate in application development. Custom coding can still be used when advanced features are required.

2.3 Emerging Trends in Low-Code Application Development:-

The important emerging trends are:

- **AI-Assisted Development:** AI helps generate code, formulas, workflows, and application components.
- **Workflow Automation:** Business processes such as approvals, notifications, and record updates can be automated.
- **Citizen Development:** Non-professional users can build applications using visual development tools.
- **Cloud-Based Development:** Applications can be developed, deployed, and managed through cloud platforms.
- **API Integration:** Applications can easily connect with databases, APIs, and external services.
- **Reusable Components:** Templates and components can be reused to reduce development effort.
- **Rapid Prototyping:** Working prototypes can be created quickly and improved based on feedback.
- **Mobile Development:** Low-code platforms support responsive and mobile applications.
- **Data Analytics:** Reports, dashboards, and visualizations can be integrated into applications.

2.4 Summary

Traditional development depends mainly on manual coding, while low-code development emphasizes visual development, automation, and reusable components. With emerging technologies such as AI, cloud computing, API integration, and analytics, low-code platforms are making application development faster and more accessible.

3. Benefits and Challenges in Low-Code Development:

3.1 Benefits

1. Faster Development

Drag-and-drop components and pre-built functionality reduce the amount of manual coding required.

2. Reduced Development

Effort Developers can use existing components, integrations, and workflows rather than implementing everything from scratch.

3. Easy Prototyping

A working prototype can be created quickly, making it easier to demonstrate and refine application requirements.

4. Increased Productivity

Developers can concentrate on complex business logic while the platform handles common application functions.

5. Citizen Development

Employees with limited programming knowledge can develop applications for simple business requirements.

6. Workflow

Automation Processes such as notifications, approvals, data validation, and record updates can be automated.

7. Easier Maintenance

Visual components and centralized platform features can simplify application maintenance.

8. Integration Support

Many low-code platforms provide ready-made connectors for databases, APIs, cloud services, and business applications.

3.2 Challenges

1. Limited Customization

Highly specialized requirements may be difficult to implement if they are not supported by the platform.

2. Vendor Lock-in

Applications can become dependent on a particular platform's features, pricing, and infrastructure.

3. Security Concerns

Applications must be configured carefully to protect sensitive information and control user access.

4. Performance Limitations

Very complex or high-scale applications may require more control than a low-code platform provides.

5. Learning the Platform

Although coding requirements are reduced, users still need to understand the platform's components, workflows, and rules.

6. Cost

Advanced features, integrations, storage, and larger numbers of users may require paid plans.

7. Dependency on Platform Updates

Changes made by the platform provider can affect application functionality.

8. Complex Applications

May Still Need Coding Low-code does not completely eliminate programming. Custom functions or advanced requirements may still require conventional code.

4. Comparison of Existing Low-Code Platforms:

Comparison of Existing Low-Code Platforms					
Feature	 Zoho Creator	 Microsoft Power Apps	 OutSystems	 Mendix	 Salesforce Platform
Developer	Zoho	Microsoft	OutSystems	Mendix	Salesforce
Development Approach	Visual + low-code	Low-code + visual development	Low-code	Low-code	Low-code
Form Building	Drag-and-drop	Visual controls	Visual UI development	Visual development	Visual development
Workflow Automation	Supported	Supported through Power Automate	Supported	Supported	Supported
Database	Built-in application database	Dataverse and connectors	Built-in/data integration options	Supported	Salesforce data platform
API Integration	Supported	Extensive connectors	Supported	Supported	Supported
Mobile Development	Supported	Supported	Supported	Supported	Supported
AI Features	AI-assisted capabilities	AI/Copilot capabilities	AI-assisted development capabilities	AI-assisted development capabilities	AI-assisted capabilities
Custom Coding	Deluge and other supported options	Power Fx and extensions	Supported	Supported	Apex and other development tools
Main Strength	Business applications and rapid development	Microsoft ecosystem integration	Enterprise application development	Enterprise low-code development	CRM and business applications
Typical Users	Businesses, developers, citizen developers	Businesses and developers	Enterprise teams	Enterprise teams	CRM/business users and developers

5. APPLICATION DEVELOPED USING ZOHOCREATOR

- Application Name: Event Registration & Participant Management
- Form Name: Event_Registration_Form
- Primary Users
 1. Students
 2. Faculty
 3. External Participants
 4. Event Organizing Committee

- Report: Event_Registration_Report

FORM:

ERP

Event Registration & Participant...

Event_Registration_Form

Event_Registration_Form

Event_Registration_Form Report

ANISHA G 24...

Event_Registration_Form

Participant_name

First Name

Last Name

Email_Address

Institution_Address

Address Line 1

Address Line 2

City / District

State / Province

Postal Code

Country

Contact_Number

+91

81234 56789

ERP

Event Registration & Participant...

Event_Registration_Form

Event_Registration_Form

Event_Registration_Form Report

ANISHA G 24...

Event_Registration_Form

Participant_name

ANISHA

G

First Name

Last Name

Email_Address

anisha.g.2024.cse@rajalakshmi.edu.in

Institution_Address

71/31 Mannarsamy Koil Street

Royapuram

Address Line 1

Address Line 2

Chennai

TamilNadu

City / District

State / Province

600013

India

Postal Code

Country

Contact_Number

+91

9876543210

Event Registration & Participant Management

Trial has expired

Upgrade

Edit this application

Help

ERP

Event Registration & Participant...

Event_Registration_Form

Event_Registration_Form

Event_Registration_Form Report

ANISHA G 24...

Event_Registration_Form

College_Organization_Name

Rajalakshmi Engineering College

Expectations_From_Event

Equipments should be in neat and clean.

Team_Size

4

Preferred_Session_Date

30-Sep-2026

Preferred_Session_Slot

22:11:30

Event_Track

Talent Hunt

Participant_Type

Student

Faculty / Staff

Professional

Event Registration & Participant Management

Trial has expired

Upgrade

Edit this application

Help

ERP

Event Registration & Participant...

Event_Registration_Form

Event_Registration_Form

Event_Registration_Form Report

ANISHA G 24...

Event_Registration_Form

Participant_Type

Student

Faculty / Staff

Professional

Research Scholar

Guest

Other

Areas_Of_Interest

Cloud Computing

Agree_To_Code_Of_Conduct

Checked (Agreed)

Choice 2

Choice 3

Submit

REPORT:

ERP

Event Registration & Participant...

Event_Registration_Form

Event_Registration_Form

Event_Registration_Form Report

ANISHA G 24...

Event Registration & Participant Management

Trial has expired Upgrade

Edit this application

Help

Event_Registration_Form Report

Participant_name

Email_Address

Institution_Address

Contact

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WORKFLOW:

Event Registration & Par ...				Design	Workflow	Settings	Upgrade	Access this application
Form workflows				Schedules	Approvals	Payments	Blueprints	Batch workflows
Functions				Search for Workflows or Acti...				New Workflow
Search for Forms				Workflow Name				Details
Forms with workflows				Status				Created On
Event_Registration_Form				Created > Successful form submission				
Send registration confirm email				1 Action				
Enabled				25-Sep-2026				

6.RESULT:

The survey and implementation of the **Event Registration & Participant Management** application demonstrated how low-code platforms can simplify application development.

The application was designed using **Zoho Creator** and uses the available Basic Fields to collect participant details, contact information, institution information,

event preferences, scheduling information, and code-of-conduct acknowledgement.

The application also demonstrates important low-code concepts such as **visual form creation, field validation, centralized data collection, report generation, and workflow automation.**